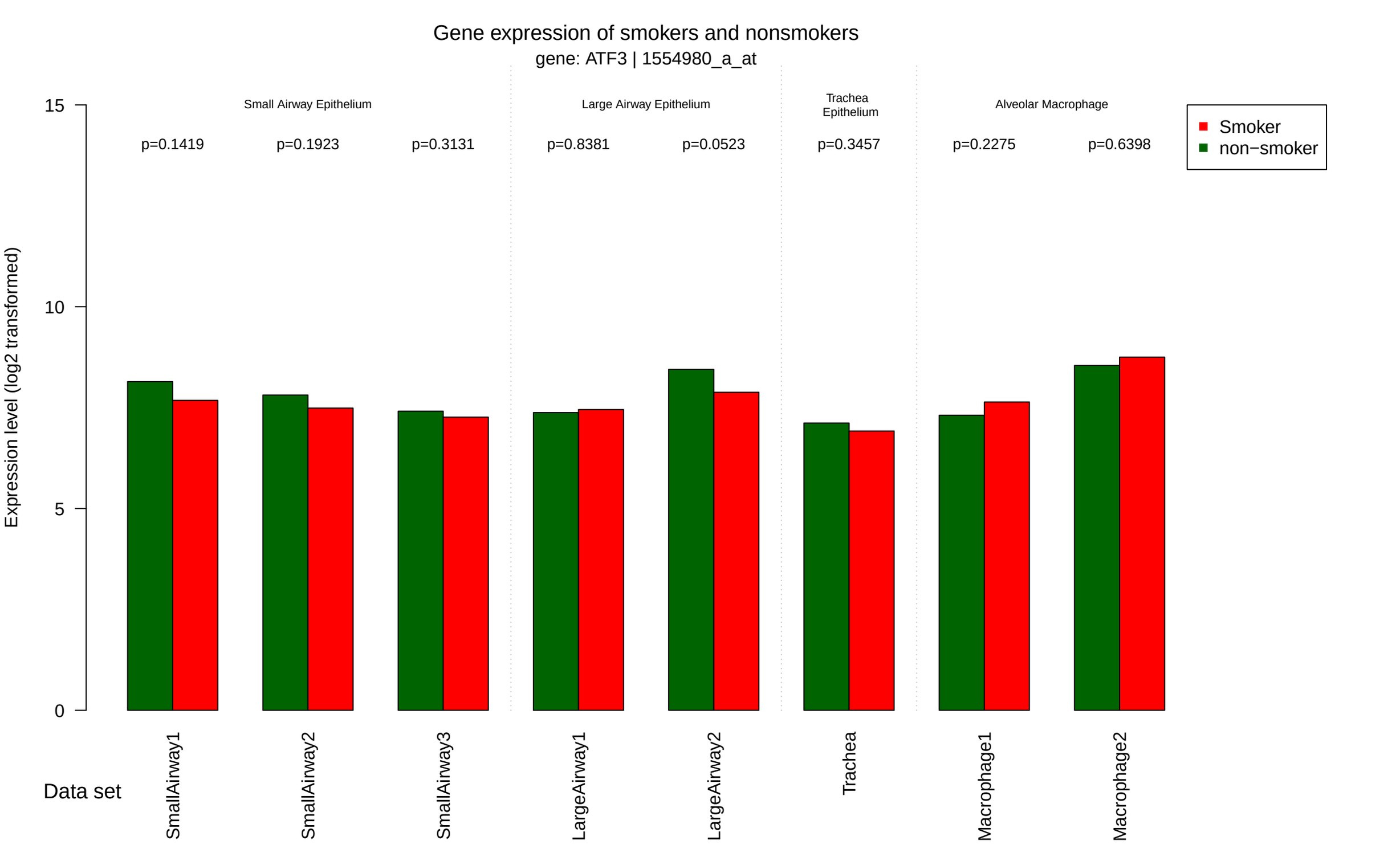
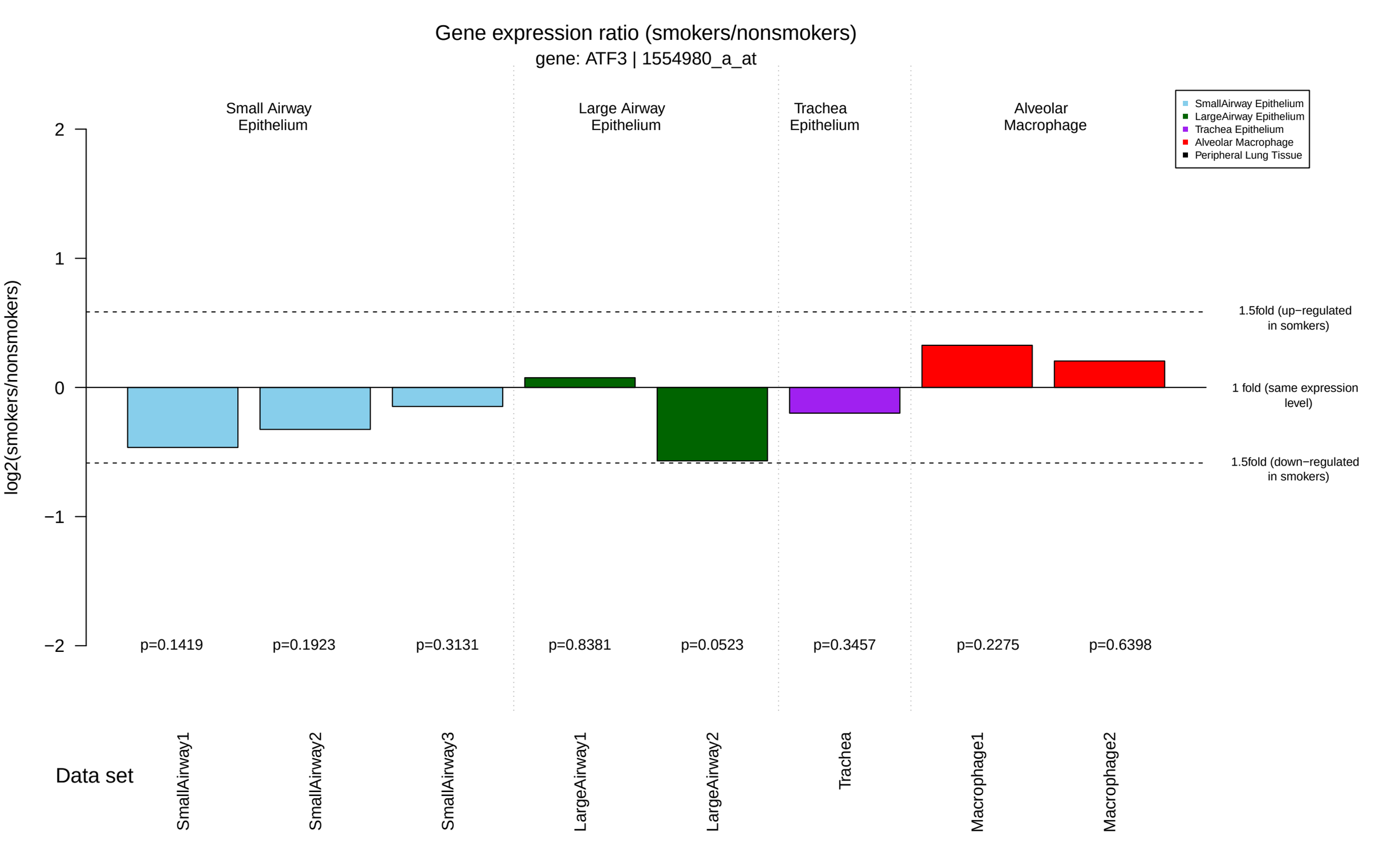
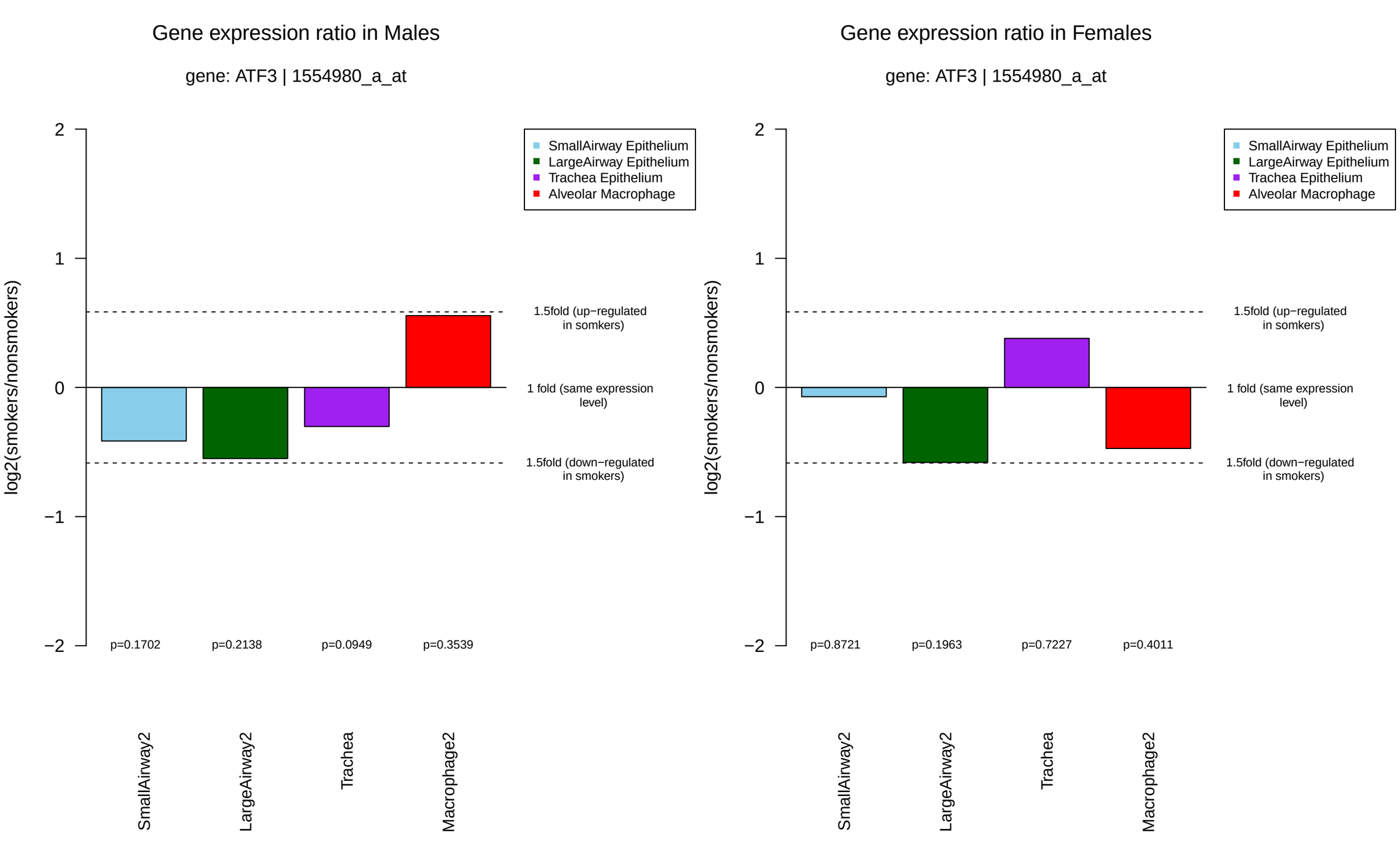
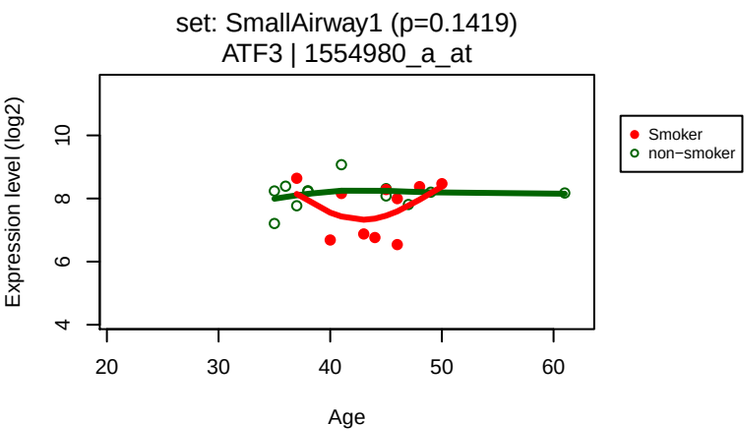
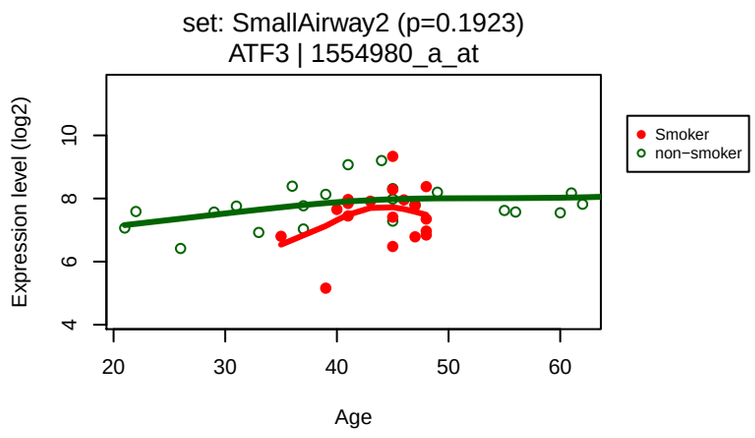
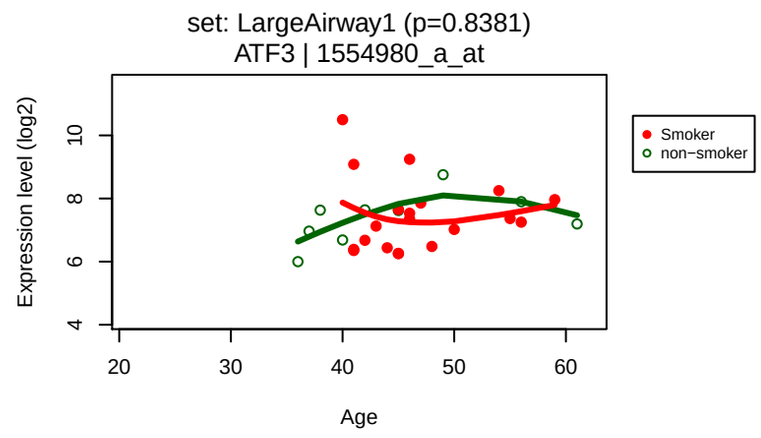

Gene expression of smokers and nonsmokers
gene: ATF3 | 1554980_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1419
p=0.1923
p=0.3131
p=0.8381
p=0.0523
p=0.3457
p=0.2275
p=0.6398
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

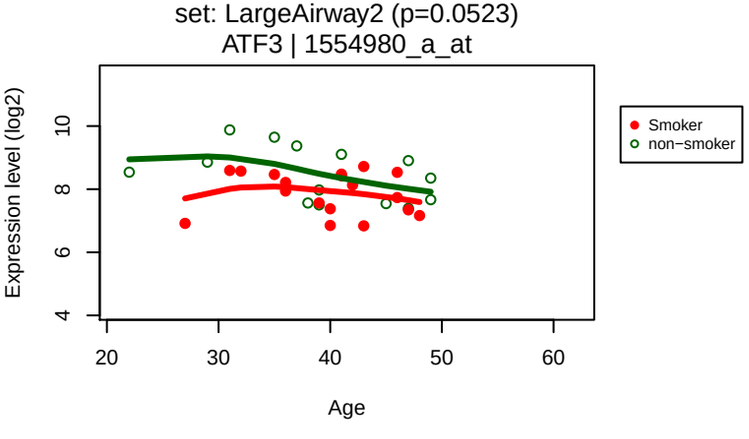

Gene expression ratio (smokers/nonsmokers)
gene: ATF3 | 1554980_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1419
p=0.1923
p=0.3131
p=0.8381
p=0.0523
p=0.3457
p=0.2275
p=0.6398
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

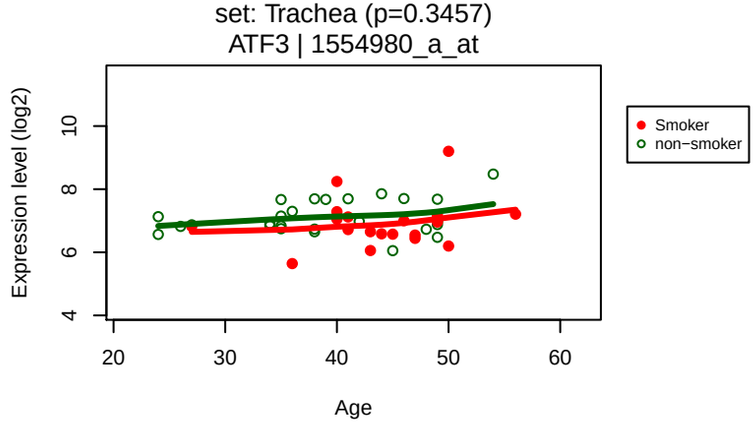

Gene expression ratio in Males
gene: ATF3 | 1554980_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1702
p=0.2138
p=0.0949
p=0.3539
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF3 | 1554980_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8721
p=0.1963
p=0.7227
p=0.4011
SmallAirway2
LargeAirway2
Trachea
Macrophage2

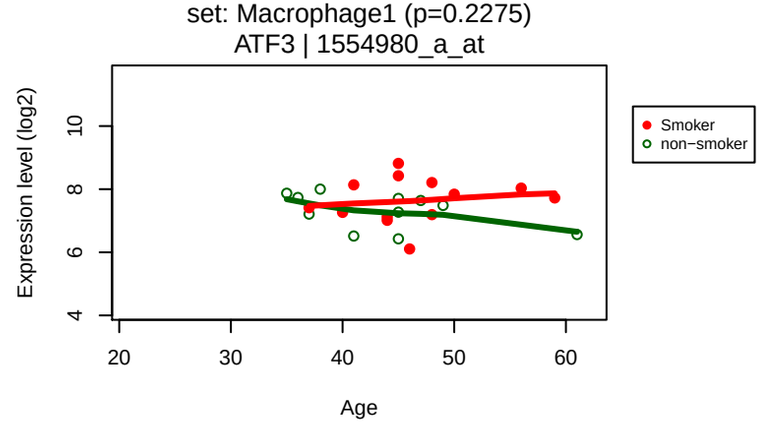

set: SmallAirway1 (p=0.1419)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker

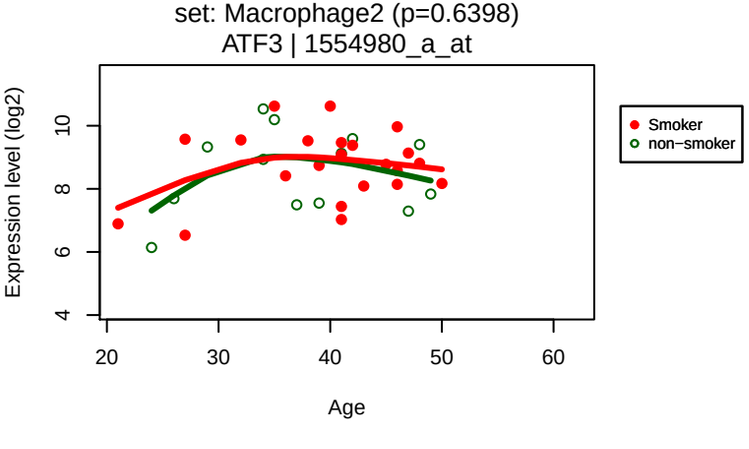

set: SmallAirway2 (p=0.1923)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker

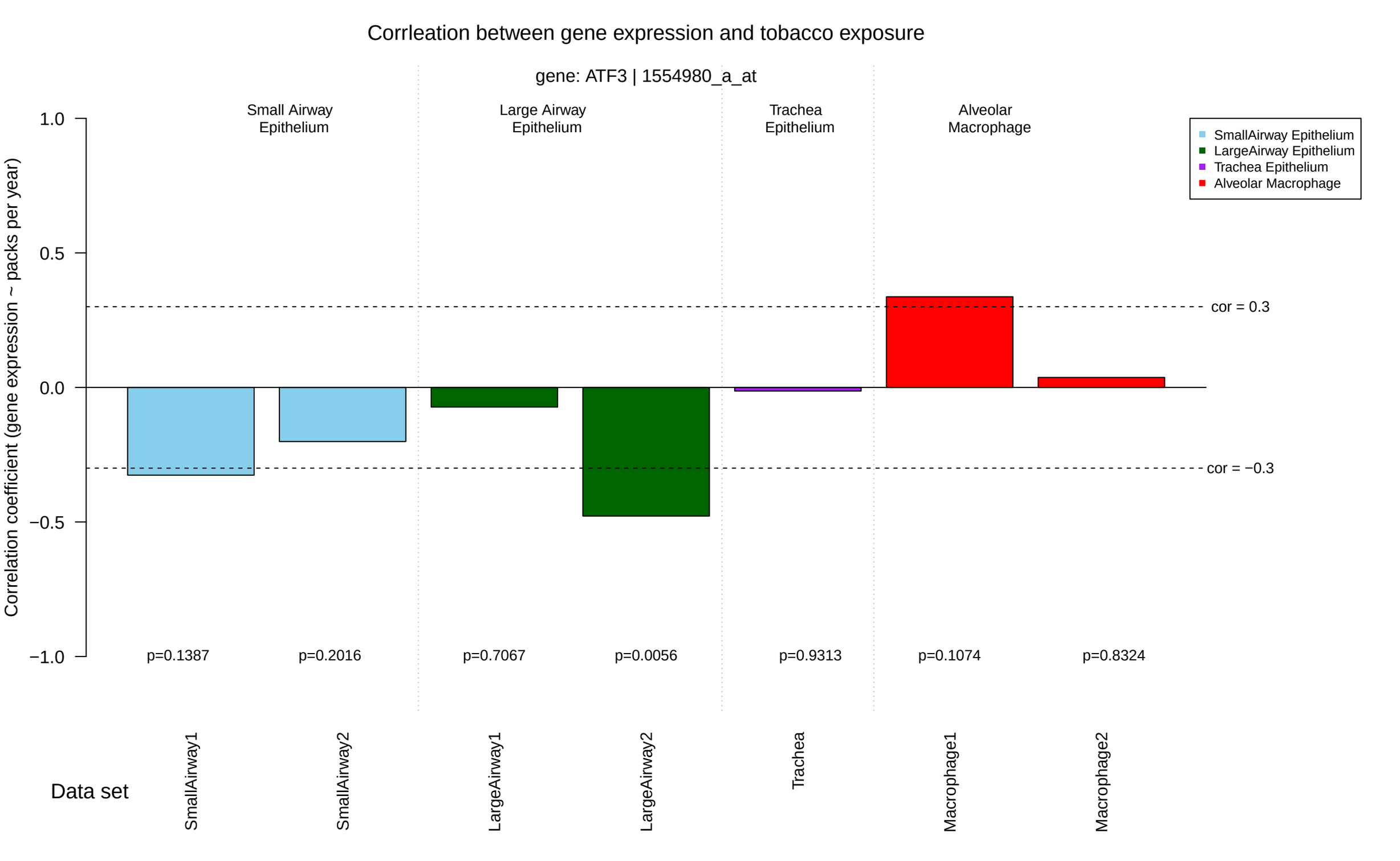

set: LargeAirway1 (p=0.8381)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0523)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3457)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2275)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6398)
ATF3 | 1554980_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF3 | 1554980_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1387
p=0.2016
p=0.7067
p=0.0056
p=0.9313
p=0.1074
p=0.8324
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2